

Georgios Palkanoglou

Contact

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Current position

Postdoctoral Researcher, TRIUMF Canada's Particle Accelerator Centre, Vancouver, Canada

Education

- 2019-2023 Ph.D. in Physics, University of Guelph
Dissertation: *Pairing in Nuclear and Cold Atomic Systems*
- 2018-2019 M.Sc. in Physics, University of Guelph
Thesis: *Approaching the Thermodynamic Limit in BCS Theory*
- 2013-2017 B.Sc. (HONOURS) in Physics; National and Kapodistrian University of Athens

Awards & Honours

- 2025-2026 Catalonia Quantum Academy Visiting Scholar Award
- 2025 Canadian Institute for Nuclear Physics support for workshop organization
- 2020 Constantine and Patricia Mavroyannis Scholarship
- 2019-2020 Dean's Graduate scholarship (University of Guelph)
- 2019 International Doctoral Training Scholarship - Ph.D.'s (University of Guelph)

Conference organizing

- 2027 Lead organizer of the ECT* workshop "*Bridging Effective Field Theory Across Subatomic Scales*"
- 2025 Lead organizer of the ESNT workshop (**with CINP support**) "*Ab initio many-body calculations: where has the pairing gone?*"
- 2025 Lead organizer of the TRIUMF workshop "*SMEFT meets ChEFT*"
- 2025 Co-organizer of the TRIUMF workshop "*PAINT'26: Progress in Ab Initio Nuclear Theory*"

Spokesperson of Collaboration

- 2025 S1936 at TRIUMF: *Beam Development of Light Lanthanides for Nuclear Structure Investigations Approaching $N = Z$*

Teaching experience

2026	Instructor for “ <i>Introduction to modern ab initio nuclear theory: a crash course</i> ”, University of Barcelona, Barcelona, Spain
2023	Graduate Teaching Assistant in Physics of Music, Department of Physics, University of Guelph
2021	Graduate Teaching Assistant in Quantum Mechanics, Department of Physics, University of Guelph
2020	Graduate Teaching Assistant in (graduate) Quantum Field Theory, Department of Physics, University of Guelph
2019-2023	Private Tutor in Undergraduate Physics
2018-2023	Graduate Teaching Assistant in Applied Physics, Department of Physics, University of Guelph
2013-2017	Private Tutor in Physics, Maths, and Geometry

Student co-Supervision

2024-2025	Christos Tsekouras, Undergraduate thesis, National and Kapodistrian University of Athens
2021-2022	Michael Stuck, Undergraduate Co-op student, University of Guelph
2022	Kareena Bhalla, Undergraduate Co-op student, University of Guelph

Publications

JOURNAL ARTICLES

2026	<i>Exploring the shell structure of trapped superfluids</i> G. Palkanoglou, R. Curry, and S. Gandolfi arXiv:2607.xxxxx
2026	<i>Interplay between Nuclear Shell Structure and Pairing around Doubly Magic ^{132}Sn</i> R. Simpson, G. Palkanoglou, E.C. Brisley, J.D. Cardona, A. Czihaly, S. Kakkar, M. Simonov, E. Taylor, C. Walls, P. Weligampola, C. Chambers, F. Maldonado Millan, A. Mollaebrabimi, D. Ray, A. Weaver, J. Yu, I. Dillmann, A. Gezerlis, G. Gwinner, A.O. Macchiavelli, S. Malbrunot-Ettenauer, M.P. Reiter, and A.A. Kwiatkowski arXiv:2607.xxxxx
2026	<i>Evidence for Mutlimodal Superfluidity of Neutrons</i> Y. Ma, G. Palkanoglou, J. Carlson, S. Gandolfi, A. Gezerlis, G. Given, A. Hicks, D. Lee, K. E. Schmidt, and J. Yu arXiv:2602.17611
2025	<i>A novel way of recasting the Bardeen-Cooper-Schrieffer gap equations</i> G. Palkanoglou and A. Gezerlis (in press at <i>Phys. Rev. C</i>) arXiv:2510.07384

- 2025 *Symmetry properties of pairing correlations in heavy deformed nuclei*
G. Palkanoglou and A. Gezerlis
[arXiv:2505.08879](#)
- 2024 *Spin-Triplet Pairing in Heavy Nuclei is Stable Against Deformation*
G. Palkanoglou, M. Stuck, and A. Gezerlis
[Phys. Rev. Lett. 134, 032501 \(2025\)](#)
[arXiv:2402.13313](#)
- 2022 *The 1S_0 pairing gap in neutron matter*
S. Gandolfi, G. Palkanoglou, J. Carlson, A. Gezerlis, and K. E. Schmidt
[Condens. Matter 2022, 7\(1\), 19](#)
[arXiv:2201.01308](#)
*Marked as a Feature Paper
- 2021 *Superfluid neutron matter with a twist*
G. Palkanoglou and A. Gezerlis
[Universe 2021, 7\(2\), 24](#)
[arXiv:2012.04663](#)
*Marked as a Feature Paper
- 2020 *From odd-even staggering to the pairing gap in neutron matter*
G. Palkanoglou, F. K. Diakonov, and A. Gezerlis
[Phys. Rev. C 102, 064324 \(2020\)](#)
[arXiv:2005.05985](#)

PROCEEDINGS

- 2019 *Effective mass and pairing gap in neutron matter*
G. Palkanoglou, N. Ismail, M. Buraczynski, and A. Gezerlis
[J. Phys.: Conf. Ser. 1643 012132 \(2020\)](#).

Invited Talks & Seminars

- 2026 *Beyond the conventional nuclear superfluidity: triplet and mixed-spin superfluids across nucleonic systems, pairing at the drip-line, and more*
University of Barcelona, Barcelona, Spain
- 2026 *Revisiting superfluidity and shell-effects at the crossroads: experiments, mean-field, and ab initio descriptions*
UPC Universitat Politècnica de Catalunya, Barcelona, Spain

- 2026 *Nuclear superfluidity and clustering, many-nucleon forces, and ML methods for ab initio nuclear theory*
Job Interview (virtual), ECT*, Trento, Italy
- 2026 *Unconventional nuclear superfluidity: from the proton to the neutron drip-line, and beyond*
Job Interview (virtual), University of Jyväskylä, Jyväskylä, Italy
- 2026 (Co-organized) *Beyond the leading three-nucleon force in light nuclei*
PAINT₂₆, Vancouver, BC, Canada
- 2025 (Organized) *Pinning the subleading three-nucleon contact interaction to light nuclei*
SMEFT meets ChEFT workshop, TRIUMF, Vancouver, BC, Canada
- 2025 (Organized) *Ab initio calculations of singlet and triplet pairing gaps in nucleonic matter*
Introductory lecture at Ab initio many-body calculations: “where has the pairing gone?”, ESNT CEA, Saclay, Paris, France
- 2025 *How to describe isoscalar pairing in heavy deformed nuclei and measure its effects*
ACOT meeting, TRIUMF, Vancouver, BC, Canada
- 2025 *Putting the subleading three nucleon force to use*
PAINT₂₅, TRIUMF, Vancouver, BC, Canada
- 2025 *S1936 Progress Report: Beam Development of Light Lanthanides for Nuclear Structure Investigations Approaching $N = Z$*
NP-EEC, TRIUMF, Vancouver, BC, Canada
- 2024 *Exotic pairing phases in nuclei and neutron matter*
T-2 Seminar, LANL, Los Alamos, NM, USA
- 2024 *Exotic nuclear superfluidity in heavy nuclei*
3rd APCTP-TRIUMF joint workshop, PKNU, Busan, South Korea
- 2024 *Nuclear superfluidity in nuclei and neutron matter*
National and Kapodistrian University of Athens, Athens, Greece
- 2024 *(Pathways to) Exotic pairing in heavy nuclei*
PAINT₂₄ - Workshop on Progress In Ab Initio Nuclear Theory, TRIUMF, Vancouver, BC, Canada
- 2023 *Novel pairing and testing EFT interactions in neutron matter and nuclei*
Michigan State University
- 2023 *Mixed-spin pairing in nuclear and cold atomic systems*
(Virtual) Warsaw University of Technology
(Job Interview)
- 2023 *Mixed-spin pairing in nuclear and cold atomic systems*
(Virtual) TRIUMF Theory Group

(Job Interview)

2023 *Representing sound: Acoustics & Phonetics*
One lecture in Phonetics
School of Languages and Literature, University of Guelph, Guelph, ON, Canada

Contributed talks

2026 *Superfluid effects in deformed heavy nuclei close to the drip-lines*
Nuclear Theory in Spain I, Universidad Complutense de Madrid, Madrid, Spain

2024 *Exotic nuclear superfluidity in heavy nuclei*
2024 CeNAM Frontiers, University of Notre Dame, South Bend, IN, USA

2024 *Exotic nuclear superfluidity in heavy nuclei*
Theory Canada 16, Western University, London, ON, Canada

2024 *Exotic nuclear superfluidity in heavy nuclei*
CAP Congress 2024, Western University, London, ON, Canada

2023 *Singlet and triplet pairing in nuclear and cold atomic systems*
2023 Canadian Association of Physicists Congress
University of New Brunswick, Fredericton, NB, Canada

2023 *Singlet and triplet pairing in nuclear and cold atomic systems*
60th Winter Nuclear and Particle Physics Conference
Banff, AB, Canada

2023 *Mixed-spin pairing in nuclear and cold atomic systems*
Toronto Ultracold Atom Network (TUCAN) meeting
University of Toronto, Toronto, ON, Canada

2022 *The S-wave pairing gap in neutron matter*
2022 Canadian Association of Physicists Congress
*Marked as the 3rd best presentation in the Division of Theoretical Physics (DTP)

2021 *Superfluid neutrons: from particles to matter*
2021 Canadian Association of Physicists (Virtual) Congress
*Tied as the 2nd best presentation in the Division of Theoretical Physics (DTP)

2021 *Superfluid neutron matter with a twist*
58th Winter Nuclear and Particle Physics (Virtual) Conference
* Marked as the 2nd best presentation in Theoretical Physics

2021 *Superfluid neutron matter with a twist*
(Virtual) Graduate Seminar Series

University of Guelph, Guelph, ON, Canada

2020 *From odd-even staggering to the pairing gap in neutron matter*
2020 Canadian Association of Physicists (Virtual) Congress

2020 *From odd-even staggering to the pairing gap in neutron matter*
57th Winter Nuclear and Particle Physics Conference
Banff, AB, Canada

2018 *A Superconducting Star*
Graduate Seminar Series
University of Guelph, Guelph, ON, Canada

Conferences & Summer schools attended

2026 ML2026 - II Machine Learning Methods for Complex and Quantum Systems, University of Barcelona, Barcelona, Spain
2026 NT.ES I - Nuclear Theory in Spain, Universidad Complutense de Madrid, Madrid, Spain
2026 CSQCD - Compact Stars in the QCD phase diagram, University of Barcelona, Barcelona, Spain
2026 PAINT₂₆ - Workshop on Progress In Ab Initio Nuclear Theory, TRIUMF, Vancouver, British Columbia, Canada
2025 SMEFT meets ChEFT, TRIUMF, Vancouver, British Columbia, Canada
2025 Ab initio many-body calculations: where has the pairing gone, ESNT, Saclay, France
2025 PAINT₂₅ - Workshop on Progress In Ab Initio Nuclear Theory, TRIUMF, Vancouver, British Columbia, Canada
2024 3rd APCTP-TRIUMF joint workshop: From Nuclei to Neutron Stars, PKNU, Busan, South Korea
2024 2024 CeNAM Frontiers, University of Notre Dame, South Bend, Indiana, USA
2024 2024 C.A.P Congress, Western University, London, Ontario, Canada
2024 Theory Canada 16, Western University, London, Ontario, Canada
2024 PAINT₂₄ - Workshop on Progress In Ab Initio Nuclear Theory, TRIUMF, Vancouver, British Columbia, Canada
2023 2023 C.A.P Congress, University of New Brunswick, Fredericton, New Brunswick, Canada
2023 60th Winter Nuclear and Particle Physics (Virtual) Conference, Banff, AB, Canada
2022 2022 C.A.P Congress, McMaster University, Hamilton, Ontario, Canada
2021 2021 C.A.P (Virtual) Congress
2021 58th Winter Nuclear and Particle Physics (Virtual) Conference
2020 2020 C.A.P (Virtual) Congress
2020 57th Winter Nuclear and Particle Physics Conference, Banff, AB, Canada
2017 The 2017 Onassis Foundation Lectures in Physics: "Quantum physics frontiers explored with cold atoms, molecules and photons", Heraklion, Greece
2017 The 2017 Petnica Summer Institute summer school on Cosmology, Petnica, Serbia
2016 The 3rd Xmas Theoretical Physics Workshop, Athens, Greece

Computing skills

LANGUAGES

Fortran, Python, Bash, MATLAB, Mathematica, C++, HTML

SOFTWARE

L^AT_EX, PLUTO, Logger Pro

Outreach & Service

- 2025 Volunteer mentor in the Early Career Researcher mentorship program, TRIUMF
- 2022 Research Information Session, University of Guelph
- 2021 Volunteer judge for the 57th Annual Canadian Undergraduate Physics Conference (CUPC)
- 2019 Volunteer for STEM Week 2019 organized by the University of Guelph, Guelph Civic Museum
- 2018 Volunteer for STEM Week 2018 organized by the University of Guelph, Guelph Civic Museum

Languages

Greek (native), English (fluent), Farsi (beginner), Spanish (beginner), German (beginner)